DAY_5

- **Inter-VLAN Routing Concepts???? Theory: Why is Inter-VLAN Routing Needed?Problem: VLANs cannot communicate with each otherSolution: Use a router or Layer 3 switchMethods of Inter-VLAN Routing:Router-on-a-Stick (RoS) (Single router interface for multiple VLANs)Layer 3 Switch Routing???? Practical: Router-on-a-Stick Configuration???? Hands-on LabEnable router sub-interfaces for VLANss**

1. On a Cisco Packet Tracer or GNS3 simulation, create three VLANs named Insta, Zomato, and Flipkart on a single switch, and assign at least one PC to each VLAN.

ANS :

- **Topology**

```
                 Router0
                  G0/0
                    |
                    | TRUNK
                    |
                  G0/1
                 Switch0
             _______|_______
             |      |      |
            PC0    PC1    PC2
        VLAN 10  VLAN 20  VLAN 30
          Insta   Zomato  Flipkart
  192.168.10.10  192.168.20.10  192.168.30.10
```

- **VLAN Details**

VLAN ID	VLAN Name	Network	Default Gateway
10	Insta	192.168.10.0/24	192.168.10.1
20	Zomato	192.168.20.0/24	192.168.20.1
30	Flipkart	192.168.30.0/24	192.168.30.1

- **Switch Configuration – Create VLANs**

```
enable
configure terminal

vlan 10
name Insta
exit

vlan 20
name Zomato
exit

vlan 30
name Flipkart
    exit
```

2. Configure a Router-on-a-Stick setup by creating sub-interfaces (e.g., GigabitEthernet0/0.10, .20, .30) on a simulated router, each with an IP address for the corresponding VLANs Insta, Zomato, and Flipkart.

ANS :

- **Configure Router-on-a-Stick**

```
enable
configure terminal

interface gigabitEthernet0/0
no shutdown
exit

! ========================
! VLAN 10 - Insta
! ========================

interface gigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit
```

```
! ==========================
! VLAN 20 - Zomato
! ==========================

interface gigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit

! ==========================
! VLAN 30 - Flipkart
! ==========================

interface gigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
exit

end
    write memory
```

3. Assign the switch port connected to the router as a trunk port, and verify trunking is active using the show interfaces trunk command. Hint: Use the switchport mode trunk command on the relevant interface.

ANS :

- **Configure the Switch Trunk Port**

```
enable
configure terminal

interface gigabitEthernet0/1
switchport mode trunk
no shutdown
exit

    end
```

- **Verify trunking:**
    - show interfaces trunk

- **Expected output:**
    - Port Mode Encapsulation Status Native vlan
      Gi0/1 on 802.1q trunking 1

    - Port Vlans allowed on trunk
      Gi0/1 1-4094

    - Port Vlans allowed and active in management domain
      Gi0/1 1,10,20,30

4. Test inter-VLAN connectivity by pinging from a PC in the Insta VLAN to a PC in the Zomato VLAN, and explain why the ping succeeds or fails based on your configuration.

ANS :

- **Test Inter-VLAN Connectivity**

  From **PC0 in VLAN 10**, open:
  Desktop → Command Prompt

  Ping PC1 in VLAN 20:
  ping 192.168.20.10

  You can also test PC2:
  ping 192.168.30.10

  Expected Result
  Reply from 192.168.20.10

- **Why Does the Ping Succeed?**

```
PC0
VLAN 10
192.168.10.10
  |
  v
Switch0
  |
  | VLAN 10 tagged traffic
  | TRUNK
  v
Router G0/0.10
192.168.10.1
  |
  | Routes traffic between networks
  v
Router G0/0.20
192.168.20.1
  |
  | VLAN 20 tagged traffic
  v
Switch0
  |
  v
PC1
VLAN 20
    192.168.20.10
```

5. Refactor your Router-on-a-Stick configuration to add a new VLAN called 'Spotify' without disrupting existing VLAN communication, and document the exact commands you used for both the switch and the router.

ANS :

- **Add New VLAN – Spotify**

VLAN ID	VLAN Name	Network	Gateway
40	Spotify	192.168.40.0/24	192.168.40.1

- **Step 1: Switch Configuration for Spotify**

Create VLAN 40:
enable
configure terminal

vlan 40
name Spotify
exit
Assume the new PC is connected to FastEthernet0/4.
Configure the port:
interface fastEthernet0/4
switchport mode access
switchport access vlan 40
no shutdown
exit
If the trunk already allows all VLANs, VLAN 40 can pass automatically.
To explicitly verify the trunk:
show interfaces trunk
If required, explicitly allow VLAN 40:
interface gigabitEthernet0/1
switchport trunk allowed vlan add 40
exit
Save:
end
write memory

- **Step 2: Configure Spotify PC**

IP Address: 192.168.40.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.40.1

- **Step 3: Add Router Sub-Interface**

On Router0:
enable
configure terminal

interface gigabitEthernet0/0.40
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
exit

end
write memory
The existing configuration remains unchanged:
G0/0.10 → Insta
G0/0.20 → Zomato
G0/0.30 → Flipkart
G0/0.40 → Spotify